BITTU KUMAR

+91-7563872977 bk183040@gmail.com bittukumar183040 BittuKumar183040

BosonQ Psi Technology Pvt. Ltd. **Jan 2023 - Present**

Full Stack Software Engineer *Bangalore, Karnataka, India*

Platform Architecture & Visualization - *React, WebGL, Three.js, Spring Boot, FastAPI, AWS S3, PVC*

- Architected and delivered a **full-stack microservices simulation platform** serving 1,000+ active users and processing 200–500 REST API requests per day.
- Engineered a **real-time 2D and 3D visualization engine** using WebGL and Three.js, reducing client-side render latency by 35% through bounding-box fitting and camera auto-framing optimizations.

Backend Development & Security - *Java, Spring Boot, FastAPI, JWT, RBAC, bcrypt, Keycloak*

- Implemented **JWT-based authentication and role-based access control (RBAC)** with bcrypt password hashing, securing multi-tenant user data across all microservices.
- Developed scalable **REST APIs** for distributed microservices using FastAPI and Spring Boot.

Frontend & Application Development - *React, SPA Architecture, TypeScript, REST Integration*

- Built a **responsive single-page application (SPA)** for a cloud-native microservices platform with scalable UI architecture.

Database Optimization - *PostgreSQL, Hibernate, ORM, Indexing*

- Optimized **PostgreSQL database schemas and ORM-based migrations**, improving query performance by 40% through indexing strategies and efficient data-access patterns.

DevOps & Deployment - *Docker, Podman, AWS, Kubernetes, CI/CD, GitOps*

- Containerized and deployed cloud-native microservices using **Docker and Podman on AWS**, with exposure to Kubernetes orchestration and GitOps-based CI/CD pipelines.

Code Quality & Design - *OOP, SOLID Principles, Code Reviews, Modular Architecture*

- Maintained high code quality through modular architecture by applying OOP and SOLID principles along with structured code review practices.

PROJECTS

3D Model Visualization | *React, WebGL, Three.js, Shaders JavaScript, File Processing,* View Github *March 2025*

- Designed and developed a **browser-based 2D and 3D model visualization tool** using React, WebGL, and Three.js, supporting multiple mesh file formats.
- Implemented advanced **scene controls** including orbit controls, axis helpers, mesh isolation, and dynamic lighting for interactive model inspection.

Windows | *React, Next, Redux, Express.js, Python, Spring Boot, Microservices, JWT , PostgreSQL* View Github *November 2025*

- Built a **pixel-perfect Windows 11 style web application** with a highly interactive UI and native-like window management behavior using React and Redux Toolkit.
- Designed a **distributed microservices architecture** comprising an Authentication service, Identity service, API Gateway, and domain-specific backend services.
- Implemented **centralized JWT validation at the API Gateway** to enforce consistent authorization and eliminate redundant token verification across services.
- Integrated **Persistent Volume Claim (PVC)** object storage for scalable and portable file storage migration.
- Delivered comprehensive **REST API documentation** using OpenAPI specification and Swagger UI.
- Improved client-side performance through **browser memory management and Redux state optimizations.**

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, SQL, HTML, CSS

Frameworks and Libraries: React, Redux, Node.js, Express.js, FastAPI, Spring Boot, Three.js, WebGL, Tailwind CSS

Databases: PostgreSQL, SQL, ORM, Database Schema Design, Query Optimization

Cloud and DevOps: AWS, AWS S3, Docker, Podman, Kubernetes, CI/CD, GitOps, SonarQube, Penetration Testing

Developer Tools: Git, GitHub, Azure Repos, VS Code, IntelliJ IDEA, PyCharm, Bruno, Swagger UI, OpenAPI

Architecture and Methodologies: Microservices, Distributed Systems, REST APIs, API Gateway, JWT, RBAC

Master of Computer Applications (MCA) *Bhopal, Madhya Pradesh, India*

Lakshmi Narain College of Technology (LNCT) *2021 - 2023*